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Roll No.

220954A

5th Sem/ Electrical
Subject : Solar Panel Installation & Maintenance

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which cell is used to convert solar energy directly into electrical energy_____.
- a) Dry Cell b) Photo electric cell
c) Battery d) None of the above
- Q.2 Which meter is used to measure the solar radiation flux_____.
- a) Pyranometer b) Sunshine Recorder
c) Anemometer d) All of the above
- Q.3 When installing a solar module on a roof, things to be-considered are?
- a) Obstructions on the roof
b) The condition of the roof.
c) Far shading
d) All of the above

- Q.4 Most widely used solar material is
- a) Arsenic b) Cadmium
 - c) Silicon d) Steel
- Q.5 To check the Solar cell temperature, use a tool known as
- a) Non contact thermometer
 - b) Digital Multimeter
 - c) Digital clamp on meter
 - d) Any of the above
- Q.6 To keep track of how much electricity of your solar panels produce and how much electricity you use from the grid, a special meter is required is known
- a) Energy meter b) Net meter
 - c) Solar meter d) Any of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand MPPT.
- Q.8 A _____ tilt allows the solar panel to self clean.
- Q.9 Tell one safety to work with PV system.
- Q.10 Expand PV.
- Q.11 Assemble of interconnected solar cell is known as solar module. (True/False)

Q.12 Name the devices which interface the solar panel and solar charge controller.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain series and parallel connection of solar cells.

Q.14 Write the difference between PV cell, PV panel, PV array, PV module.

Q.15 Describe occupational health and safety assessment during solar installation.

Q.16 Explain how maintaining the work area safe and secure during PV installation.

Q.17 What are significance of voltage, amperes and watt is solar cells?

Q.18 Write the merit and demerit of off-grid PV power generation system.

Q.19 Describe Solar Tracking System with neat diagram.

Q.20 list & Explain tools for installation and maintaining photovoltaic Power system.

Q.21 Explain, how build up the team coordination between colleagues at workplace.

Q.22 Write about on grid Solar power Generation system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the different types of solar panel mounting structures and their merits and demerits.
- Q.24 Explain the steps for designing and evaluation of various parameters of the solar PV system.
- Q.25 Draw the basic block diagram for solar photovoltaic power plant. Explain the each block in details.